

## SAFETY DATA SHEET

Creation Date 26-September-2009

Revision Date 27-March-2024

Revision Number 3

### 1. Identification

|                             |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| <b>Product Name</b>         | <b>Hexaamminecobalt (III) chloride</b>                                                  |
| <b>Cat No. :</b>            | <b>A15470</b>                                                                           |
| <b>CAS-No</b>               | 10534-89-1                                                                              |
| <b>Synonyms</b>             | Cobalt hexammine chloride; Hexaammine cobalt(III)chloride; Hexaamminecobalt trichloride |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                   |
| <b>Uses advised against</b> | Food, drug, pesticide or biocidal product use.                                          |

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                  |             |
|----------------------------------|-------------|
| <b>Respiratory Sensitization</b> | Category 1  |
| <b>Skin Sensitization</b>        | Category 1  |
| <b>Carcinogenicity</b>           | Category 1B |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May cause an allergic skin reaction

May cause allergy or asthma symptoms or breathing difficulties if inhaled

May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear respiratory protection

**Response**

IF exposed or concerned: Get medical advice/attention  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
If experiencing respiratory symptoms: Call a POISON CENTER/doctor  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                        | CAS-No     | Weight % |
|--------------------------------------------------|------------|----------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | 99.99    |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                                                                                                                                                                                                             |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                  |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                       |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                              |
| <b>Most important symptoms/effects</b> | May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                              |

### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | Not applicable                                                                        |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Ammonia. Cobalt oxides. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                     |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong oxidizing agents. Halogens. Strong acids.                              |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                                        | Alberta                     | British Columbia            | Ontario TWAEV               | Quebec                      | ACGIH TLV                   | OSHA PEL | NIOSH |
|--------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|----------|-------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> |          |       |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact,

and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Powder Solid             |
| <b>Appearance</b>                             | Orange, Brown            |
| <b>Odor</b>                                   | Odorless                 |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | 4 - 6 @ 25°C 26,7 g/l    |
| <b>Melting Point/Range</b>                    | 217                      |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| Upper                                         | No data available        |
| Lower                                         | No data available        |
| <b>Vapor Pressure</b>                         | No information available |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Specific Gravity</b>                       | 1.7000                   |
| <b>Solubility</b>                             | Soluble in water         |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | Not applicable           |
| <b>Decomposition Temperature</b>              | No information available |
| <b>Viscosity</b>                              | Not applicable           |
| <b>Molecular Formula</b>                      | H18 Cl3 Co N6            |

Molecular Weight

267.47

## 10. Stability and reactivity

|                                  |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                           |
| Stability                        | Stable under recommended storage conditions.                         |
| Conditions to Avoid              | Avoid dust formation. Incompatible products.                         |
| Incompatible Materials           | Strong oxidizing agents, Halogens, Strong acids                      |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Ammonia, Cobalt oxides, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                             |
| Hazardous Reactions              | None under normal processing.                                        |

## 11. Toxicological information

### Acute Toxicity

### Product Information

### Component Information

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                         |
|-----------------|-----------------------------------------|
| Irritation      | No information available                |
| Sensitization   | May cause sensitization by skin contact |
| Carcinogenicity | .                                       |

| Component                                        | CAS-No     | IARC       | NTP                    | ACGIH | OSHA       | Mexico     |
|--------------------------------------------------|------------|------------|------------------------|-------|------------|------------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | Not listed | Reasonably Anticipated | A3    | Not listed | Not listed |

NTP: (National Toxicity Program)

NTP: (National Toxicity Program)

Known - Known Carcinogen

Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

|                          |                           |
|--------------------------|---------------------------|
| Mutagenic Effects        | No information available  |
| Reproductive Effects     | No information available. |
| Developmental Effects    | No information available. |
| Teratogenicity           | No information available. |
| STOT - single exposure   | None known                |
| STOT - repeated exposure | None known                |
| Aspiration hazard        | No information available  |

Symptoms / effects, both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

**Persistence and Degradability** based on information available. May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component                                        | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | X   | -    | X    | ACTIVE                                        | 234-103-9 | -      | -   |

| Component                                        | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | -     | KE-18384 | -    | X    | X    | X    | -     | -     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|--------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Cobalt(3+), hexaammine-, | Part 1, Group B Substance                            |                                                                            |                                           |

|                         |  |  |  |
|-------------------------|--|--|--|
| trichloride, (OC-6-11)- |  |  |  |
|-------------------------|--|--|--|

**Other International Regulations**

Authorisation/Restrictions according to EU REACH Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                        | CAS-No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                                        | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cobalt(3+), hexaammine-, trichloride, (OC-6-11)- | 10534-89-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Product Safety Department  
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**Creation Date** 26-September-2009  
**Revision Date** 27-March-2024  
**Print Date** 27-March-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**